

JEE MAIN CRASH COURSE

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- ✓ **LATEST NTA PATTERN**
- ✓ **CONCEPT VIDEOS**
- ✓ **LIVE DOUBT SESSIONS**
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Designed by
Math Maestro

Anup Sir



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JEE Mains 2020 Jan Chapter wise Question Bank

Point and Straight Line

Q1

If $A(1, 1)$, $B(6, 5)$, $C\left(\frac{3}{2}, 2\right)$ are vertices of $\triangle ABC$. A point P is such that area of $\triangle PAB$, $\triangle PAC$, $\triangle PBC$ are equal, also $Q\left(\frac{-7}{6}, \frac{-1}{3}\right)$, then length of PQ is

यदि $A(1, 1)$, $B(6, 5)$, $C\left(\frac{3}{2}, 2\right)$ त्रिभुज $\triangle ABC$ के शीर्ष है एक बिन्दु P इस प्रकार है कि $\triangle PAB$, $\triangle PAC$, $\triangle PBC$ के क्षेत्रफल बराबर है तथा $Q\left(\frac{-7}{6}, \frac{-1}{3}\right)$ तब PQ की लम्बाई है—

(1) 2

(2) 3

(3) 4

(4) 5

7th Jan Morning

Sol

(4)

P will be centroid of $\triangle ABC$

P त्रिभुज $\triangle ABC$ का केन्द्रक होगा

$$P\left(\frac{17}{6}, \frac{8}{3}\right) \Rightarrow PQ = \sqrt{\left(\frac{24}{6}\right)^2 + \left(\frac{9}{3}\right)^2} = 5$$

02

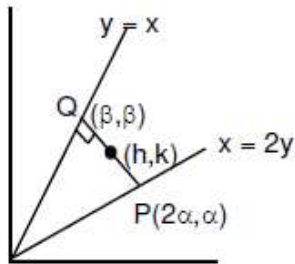
From any point P on the line $x = 2y$ perpendicular is drawn on $y = x$. Let foot of perpendicular is Q . Find the locus of mid point of PQ .

रेखा $x = 2y$ पर स्थित किसी बिन्दु P से रेखा $y = x$ पर लम्ब खींचा जाता है। माना Q लम्बपाद है तब PQ के मध्यबिन्दु का बिन्दुपथ ज्ञात करो—

(1) $2x = 3y$ (2) $5x = 7y$ (3) $3x = 2y$ (4) $7x = 5y$ 7th Jan Evening

Sol

(2)



slope of PQ की प्रवणता = $\frac{k - \alpha}{h - 2\alpha} = -1$

$$\Rightarrow k - \alpha = -h + 2\alpha$$

$$\Rightarrow \alpha = \frac{h+k}{3} \quad \dots\dots(1)$$

Also तथा $2h = 2\alpha + \beta$

$$2k = \alpha + \beta$$

$$\Rightarrow 2h = \alpha + 2k$$

$$\Rightarrow \alpha = 2h - 2k \quad \dots\dots(2)$$

from (1) और & (2) से

$$\frac{h+k}{3} = 2(h-k)$$

so locus is अतः बिन्दुपथ $6x - 6y = x + y \Rightarrow 5x = 7y$

03

Let P be a point on $x^2 = 4y$. The segment joining A (0, -1) and P is divided by point Q in the ratio 1:2, then locus of point Q is

माना $x^2 = 4y$ पर एक बिन्दु P है। बिन्दु A (0, -1) एवं P को मिलाने वाला रेखाखण्ड बिन्दु Q द्वारा 1:2 में विभाजित होता है, तब Q का बिन्दु पथ है:

- (1) $9x^2 = 3y + 2$ (2) $9x^2 = 12y + 8$ (3) $9y^2 = 12x + 8$ (4) $9y^2 = 3x + 2$

8th Jan Morning

Sol

(2)

Let point P be $(2t, t^2)$ and Q be (h, k) .माना बिन्दु P $(2t, t^2)$ तथा Q (h, k) है।

$$h = \frac{2t}{3}, k = \frac{-2+t^2}{3}$$

$$\text{Hence locus is } 3k + 2 = \left(\frac{3h}{2}\right)^2 \Rightarrow 9x^2 = 12y + 8$$

$$\text{अतः बिन्दु पथ } 3k + 2 = \left(\frac{3h}{2}\right)^2 \Rightarrow 9x^2 = 12y + 8$$

04

Let ABC is a triangle whose vertices are $A(1, -1)$, $B(0, 2)$, $C(x', y')$ and area of $\triangle ABC$ is 5 and $C(x', y')$ lie on $3x + y - 4\lambda = 0$, then

माना त्रिभुज ABC एक त्रिभुज है जिसके शीर्ष $A(1, -1)$, $B(0, 2)$, $C(x', y')$ है तथा त्रिभुज ABC का क्षेत्रफल 5 है तथा $C(x', y')$ रेखा $3x + y - 4\lambda = 0$ पर स्थित है, तब

(1) $\lambda = 3$

(2) $\lambda = -3$

(3) $\lambda = 4$

(4) $\lambda = 2$

8th Jan Morning

Sol

(1)

$$D = \frac{1}{2} \begin{vmatrix} 0 & 2 & 1 \\ 1 & -1 & 1 \\ x' & y' & 1 \end{vmatrix}$$

$$-2(1 - x') + (y' + x') = \pm 10$$

$$-2 + 2x' + y' + x' = \pm 10$$

$$3x' + y' = 12 \text{ or } 3x' + y' = -8$$

$$\lambda = 3, -2$$

05

Normal at $(2, 2)$ to curve $x^2 + 2xy - 3y^2 = 0$ is L. Then perpendicular distance from origin to line L is

वक्र $x^2 + 2xy - 3y^2 = 0$ के बिंदु $(2, 2)$ पर अभिलम्ब L है, तब मूल बिंदु से रेखा L पर लम्ब की लंबाई है—

(1) $4\sqrt{2}$

(2) 2

(3) $2\sqrt{2}$

(4) 4

8th Jan Evening

Sol

(3)

$$x^2 + 2xy - 3y^2 = 0$$

$$x^2 + 3xy - xy - 3y^2 = 0$$

$$(x - y)(x + 3y) = 0$$

$$x - y = 0 \quad x + 3y = 0$$

(2, 2) satisfy $x - y = 0$ संतुष्ट करता है

Normal अभिलम्ब :

$$x + y = \lambda$$

$$\lambda = 4$$

Hence इस प्रकार $x + y = 4$

$$\text{perpendicular distance from origin} = \left| \frac{0+0-4}{\sqrt{2}} \right| = 2\sqrt{2}$$

06

$[\text{PdFClBrI}]^{2-}$ Number of Geometrical Isomers = n. For $[\text{Fe}(\text{CN})_6]^{n-6}$, Determine the spin only magnetic moment and CFSE (Ignore the pairing energy)

- (1) 1.73 B.M., $-2\Delta_0$ (2) 2.84 B.M., $-1.6\Delta_0$ (3) 0, $-1.6\Delta_0$ (4) 5.92 B.M., $-2.4\Delta_0$

9th Jan Morning

Sol

(1)

Number of Geometrical Isomers in square planar $[\text{PdFClBrI}]^{2-}$ are = 3

Hence, $n = 3$

$[\text{Fe}(\text{CN})_6]^{3-}$

$\text{Fe}^{3+} = 3d^5$, According to CFT configuration is $t_{2g}^{221} e_g^{00}$

$$\mu = \sqrt{n(n+2)} = 1.73 \text{ B.M.}$$

$$\text{CFSE} = -0.4\Delta_0 \times n_{t_{2g}} + 0.6\Delta_0 \times n_{e_g}$$

$$= -0.4\Delta_0 \times 5 = -2.0\Delta_0$$

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